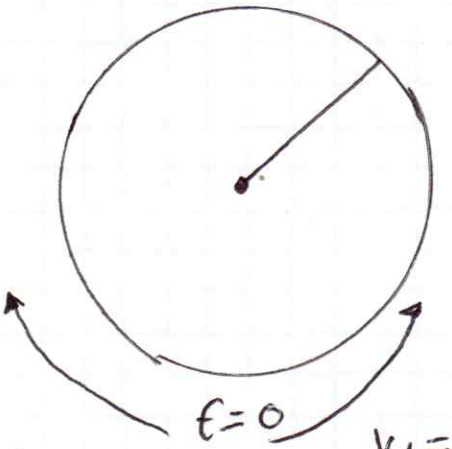
$L = 2\pi r$

$d = vt$

$d_1 = v_1 t$

$d_2 = v_2 t$




$v_2 = 5.50 \frac{m}{s}$ $\quad v_1 = 6.20 \frac{m}{s}$

$d_1 + d_2 = v_1 t + v_2 t$

$200m = (v_1 + v_2)t$

$t = \frac{200m}{v_1 + v_2} = \frac{200}{6.20 \frac{m}{s} + 5.50}$

$= 17.945 = 17.09s$

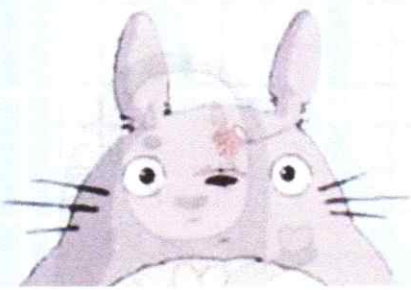
$d_1 = 6.20 \frac{m}{s} \times 17.0945 = 105.983 m$

$d_1 = 105.98 m$

$\cong 106 m$

$d_2 = 200m - 106 m$

$= 94m$



2

Particula A:

posicion inicial $(0,0)$ m

- velocidad $v_a = (12, -3)$ m/s

Particula B

- Posicion inicial $(32, -8)$ m
- velocidad $v_B = (-4, 1)$ m/s

Particula A

$r_a(t) = (0,0) + (12, -3)t$

- $x_a(t) = 12t$
  $y_a(t) = -3t$

particula B

$r_a(t) = (0,0) + (12, -3)t$

$x_B(t) = 32 - 4t$
$y_B(t) - 8 + t$

$x_B(2) = 32 - 4(2) = 3 - 2 = 24$ m

$y_B(2) = -8 + 1(2) = -8 + 2 = -6$ m

si se cruzan tiempo encuentro t=

posicion de encuentro (24-6

Particula A: $x_a(t) = 12t$, $y_a(t) = -3t$

Particula B: $x_B(t) = 32 - 4t$ $y_B(t) = -8 + t$

Formula $\vec{v} = (x\text{-final} - x\text{-inicial}) / (t\text{-final} - t\text{inicial})$

(3)

(a) De 0 a 2 s:

$\vec{v} = (8-0)/(2-0) = 4 m/s$

(b) De 0 a 4 s:

$\vec{v} = (8-0)/(4-0) = 2 m/s$

c) de 2 s. a 4 s

$\vec{v} = (8-8)/(4-2) = 0 m/s$

d) De 4 s a 7 s:

$\vec{v} = (-6-8)/(7-4) = -14/3 = -4.67 m/s$

e) De 0 a 8 s:

$\vec{v} = (-6-0)/(8-0) = -0.75 m/s$

5) 5

a 20000m a Km y mm

- 1km = 1000 m

1 m = 1000mm

- 20000 m / 1000 = 20 km

A milimetros = 20000 m x 1000 = 20.000,000 mm

(b) 300000000 µs a ms y s

1 s = 1.000,000 µs

- 1 s = 1000

a mili segundos 300000000 µs / 1000 = 300.000

A segundos 300000000 µs / 1,000,000 = 300 s

(E) 200 MB a B

- 1 MB = 1.000,000 B

- 200 MB x 1.000,000 = 200,000,000 B

(d) 973 g a kg y mg

1 kg = 1000 g

1 g = 1000 mg

A kilogramos 973 g / 1000 = 0.973 kg

A miligramos 973 g x 1000 = 973.000 mg

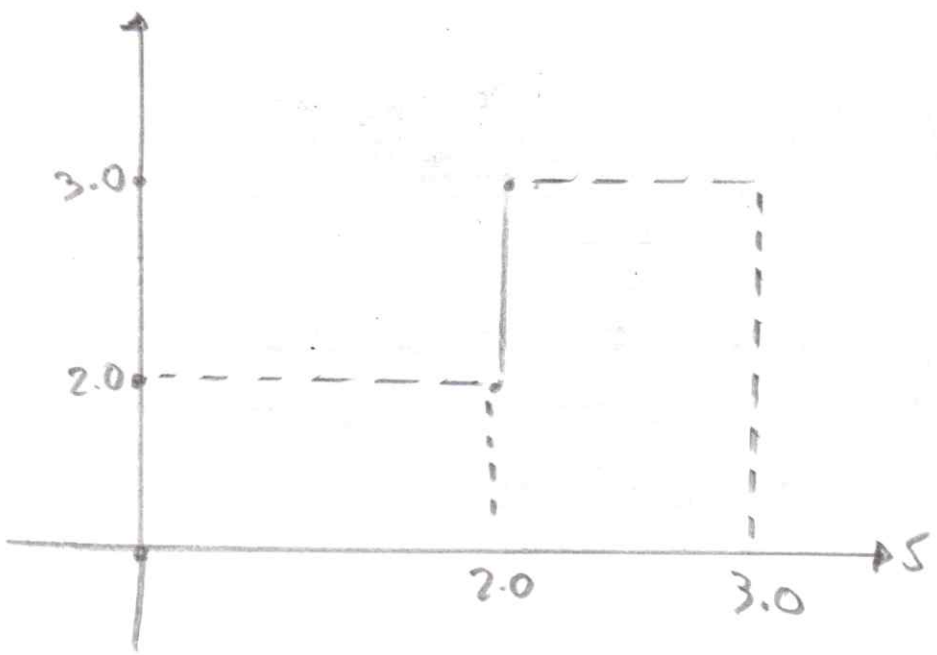
7

$$\vec{V} = \frac{\Delta x}{\Delta t} = \frac{x2 - x1}{t}$$

$d1 = vi\,ti = 2.0\frac{m}{s} \times 2.0s = 4.00m$

$d2 = v2\,t2 = 23.0\frac{m}{s} \times 1.0s = 23.00m$

$V = \text{Rapidez} = \frac{27m}{3s} = 9\frac{m}{s} \approx 9\frac{m}{s}$

$\vec{V}\ \frac{\Delta x}{\Delta t} = \frac{x^2 - y_1}{t2 - t1} = \frac{27m - 0m}{3.0s - 0s} = 9\frac{m}{s}$

y = m/s